

CS & IT ENGINEERING

DISCRETE MATHS
COMBINATORICS



Lecture No. 10



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TOPICS TO BE COVERED

01 Recurrence Relation

02 Homogenous equation

03 Non homogenous equation

COMBINATORICS



Recurrence Relation :

Colony there are some no. of bacteria.

at time 0, 5 bacteria are present
then at time 100, what will be total no. of bacteria, if bacteria are
2 times more what it was present previously?

a_0 : at time 0 $\rightarrow$ 5 bacteria.

a_1 : $\text{---} 1 \text{---} 1 \rightarrow 2 \times 5 \text{---}$

a_2 : $\text{---} 1 \text{---} 2 \rightarrow 2 \times (2 \times 5) \text{---}$

COMBINATORICS



$$a_0 \quad a_1 \quad a_2 \quad a_3 \quad \dots \quad a_{97} \quad a_{98} \quad a_{99} \quad \underline{a_{100}}$$

$$a_1 = 2 \cdot a_0$$

$$\text{initial condition} = a_0 = 5.$$

$$\underline{a_2} = 2 \cdot a_1$$

$$a_3 = 2 \cdot a_2$$

$$= 2(2 \cdot a_1)$$

$$= (2 \cdot 2(2 \cdot a_0))$$

$$a_3 = 2^3 \cdot a_0$$

$$a_{100} = 2^{100} \cdot a_0$$

COMBINATORICS

$$a_0 = 5 \text{ (initial condition)}$$

$$a_n = 2 \cdot a_{n-1}$$

↓

$$a_n = 2^n \cdot a_0 \text{ (solution)}$$

Type-1:



$$a_0 = d \text{ (initial condition)}$$

$$a_n = x \cdot a_{n-1} \text{ (problem)}$$

Solution:

$$a_n = x^n \cdot d$$

COMBINATORICS



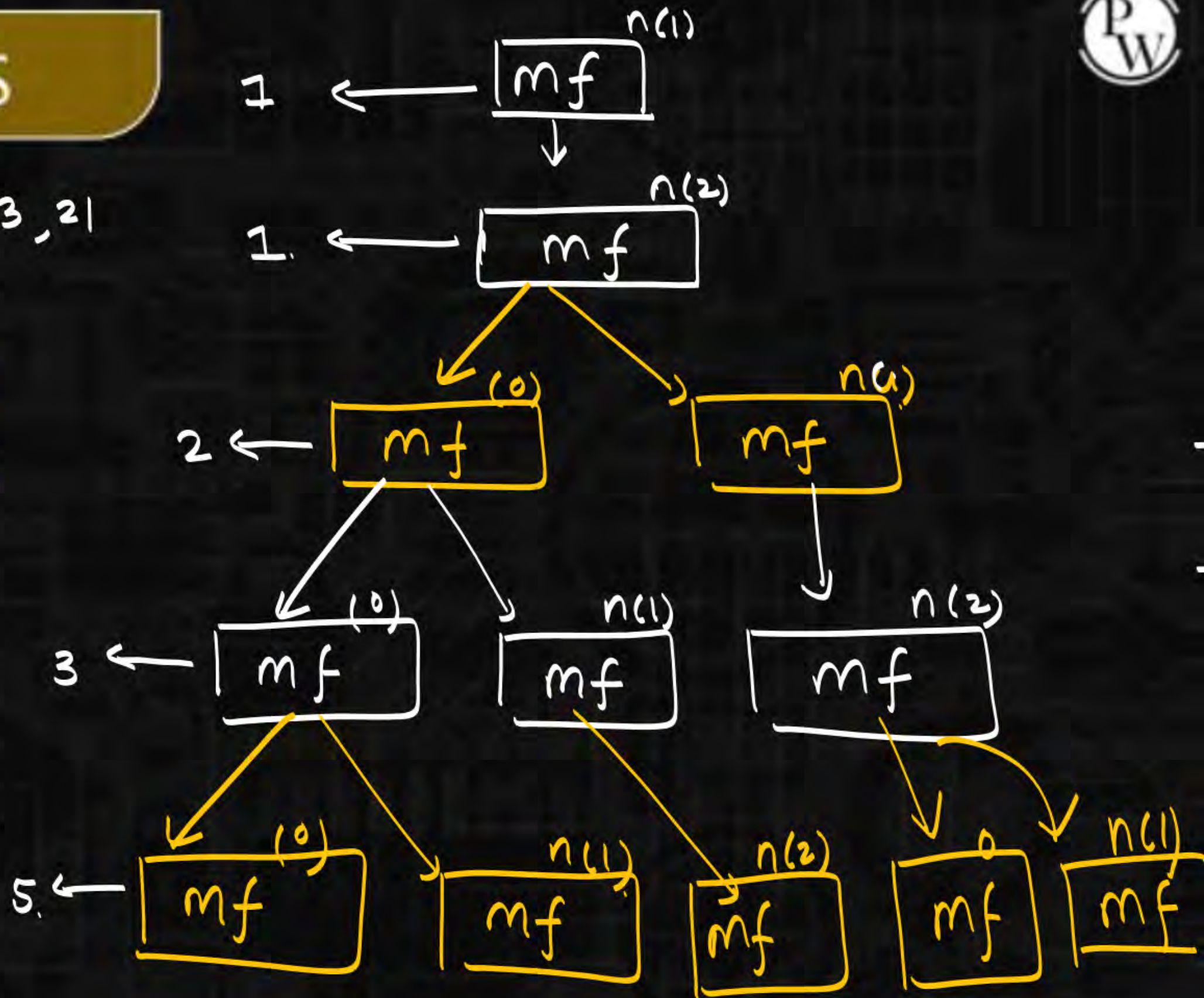
new pair $\rightarrow$ will produce pair after 2 months.

Old pair $\rightarrow$ produce every month.

new pair...

what will be total pairs after 10 months?

mf

$$a_n = a_{n-1} + a_{n-2}$$


COMBINATORICS

$$G(x) = \sum a_n \cdot x^n$$



$$a_{n+2} = 3a_{n+1} - 2a_n$$

$$a_0 = 0 \quad a_1 = 1 \quad n \geq 0$$

$$\times \sum x^{n+2}$$

$$\sum a_{n+2} x^{n+2} = 3 \sum a_{n+1} x^{n+2} - 2 \sum a_n x^{n+2}$$

$$\sum_{n \geq 0} a_{n+2} x^{n+2} = 3x \sum_{n \geq 0} a_{n+1} x^{n+1} - 2x^2 \sum_{n \geq 0} a_n x^n$$

$$G(x) - \underline{a_0 x^0} - \underline{a_1 x^1} = 3x(G(x) - \underline{a_0 x^0}) - 2x^2 \underline{G(x)}$$

COMBINATORICS



$$\sum_{n \geq 0} a_n \cdot x^n = a_0 x^0 + a_1 x^1 + \boxed{a_2 x^2 + a_3 x^3 \dots} = G(x)$$

$$\sum_{n \geq 0} a_{n+2} x^{n+2} = \boxed{a_2 x^2 + a_3 x^3 \dots \infty} = G(x) - a_0 x^0 - a_1 x^1$$

$$\left. \begin{aligned} a_0 x^0 + a_1 x^1 + \boxed{a_2 x^2 + a_3 x^3 \dots \infty} &= G(x) \\ \boxed{a_2 x^2 + a_3 x^3 \dots \infty} &= G(x) - a_0 x^0 - a_1 x^1 \end{aligned} \right\}$$

COMBINATORICS

$$G(x) = \sum a_n \cdot x^n.$$



$$(G(x) - 0 \cdot x^0 - 1 \cdot x) = 3x(G(x) - 0 \cdot x^0) - 2x^2 G(x) \quad a_0 = 0 \quad n \geq 0.$$

$$G(x) - x = 3xG(x) - 2x^2G(x)$$

$$G(x) - 3xG(x) + 2x^2G(x) = x.$$

$$G(x)[1 - 3x + 2x^2] = x.$$

$$G(x) = \frac{x}{1 - 3x + 2x^2}.$$

COMBINATORICS



$$G(x) = \frac{x}{1-3x+2x^2} = \frac{x}{(1-x)(1-2x)} = \frac{A}{1-x} + \frac{B}{1-2x}$$

$$= \frac{A(1-2x) + B(1-x)}{(1-x)(1-2x)}$$

$$x = A - 2Ax + B - Bx$$

$$= (A+B) + (-2A-B)x$$

$$\begin{aligned} -2A - B &= 1 \\ A + B &= 0 \end{aligned}$$

$$1 - 3x + 2x^2$$

$$1 - x - 2x + 2x^2$$

$$1(1-x) - 2x(1-x)$$

$$(1-x)(1-2x)$$

$$-2A - (-A) = 1$$

$$-A = 1$$

$$\begin{aligned} A &= -1 \\ B &= 1 \end{aligned}$$

COMBINATORICS



$$A = -1$$
$$B = 1$$

$$G(x) = \frac{A}{1-x} + \frac{B}{1-2x}$$

$$G(x) = \frac{-1}{1-x} + \frac{1}{1-2x}$$

$$G(x) = -\sum x^n + \sum 2^n x^n$$

$$\sum a_n x^n = -\sum x^n + \sum 2^n x^n$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \dots$$

$$= \sum x^n$$

$$\frac{1}{1-2x} = 1 + 2x + (2x)^2 + (2x)^3 + \dots$$

$$= \sum 2^n x^n$$

$$a_n = 2^n - 1$$

COMBINATORICS



$$a_{n+2} = 3a_{n+1} - 2a_n \quad n \geq 0 \quad a_0 = 0 \quad a_1 = 1.$$

$$1. \quad x^2 = 3x - 2$$

$$x^2 - 3x + 2 = 0$$

$$\boxed{x = 1, 2}$$

$$C.E: a_n = R_1^n \cdot c_1 + R_2^n \cdot c_2$$

$$a_n = 1^n \cdot c_1 + 2^n \cdot c_2$$

$$a_n = 2^n - 1$$

$$n=0$$

$$a_0 = 1^0 c_1 + 2^0 c_2$$

$$0 = c_1 + c_2$$

$$n=1$$

$$a_1 = 1^1 c_1 + 2^1 c_2$$

$$1 = c_1 + 2c_2$$

$$c_1 = -1 \quad c_2 = 1$$

COMBINATORICS



$$a_n = \frac{1}{5} \cdot 7^n + \frac{4}{5} 2^n = 4.$$

$$a_n = 9a_{n-1} - 14a_{n-2} \quad \boxed{a_0 = 1 \quad a_1 = 3}$$

$$x^2 = 9x - 14$$

$$x^2 - 9x + 14 = 0$$

$$x = 7, 2$$

$$CE: a_n = r_1^n c_1 + r_2^n c_2$$

$$a_n = 7^n c_1 + 2^n c_2$$

$$n=0$$

$$a_0 = c_1 + c_2$$

$$1 = c_1 + c_2$$

$$n=1$$

$$a_1 = 7c_1 + 2c_2$$

$$c_1 = 1/5$$

$$c_2 = 4/5$$

$$7c_1 + 2c_2 = 3$$

$$2c_1 + c_2 = 2$$

$$5c_1 = 1$$

$$c_1 + c_2 = 1$$

$$c_2 = 1 - \frac{1}{5} = \frac{4}{5}$$

COMBINATORICS



$$a_n = 7^n \cdot \frac{1}{5} + 2^n \cdot \frac{4}{5} = \frac{1}{5} (7^n + 4 \cdot 2^n)$$

$$\begin{aligned} a_0 &= 1. \\ a_1 &= 3. \end{aligned}$$

$$n=0$$

$$a_0 = \frac{1}{5} (7^0 + 4 \cdot 2^0)$$

$$= \frac{1}{5} (1 + 4 \cdot 1) = \frac{5}{5} = 1.$$

$$a_0 = 1$$

$$n=1.$$

$$a_1 = \frac{1}{5} (7 + 4 \cdot 2) = \frac{15}{5} = 3.$$

COMBINATORICS



Type-1: $a_0 = d$

$$a_n = x a_{n-1}$$

Solution: $a_n = x^n d$

Type-3: $a_n = x a_{n-1} + y a_{n-2}$

Roots: r, R

CE: $a_n = R^n c_1 + n \cdot R^n c_2$

Type-2: $a_n = x a_{n-1} + y a_{n-2}$

Roots: R_1, R_2

CE: $a_n = R_1^n c_1 + R_2^n c_2$

$n=0$

$n=1$

COMBINATORICS



$$a_{n+2} = 6a_{n+1} - 9a_n$$

$$x^2 = 6x - 9$$

$$x^2 - 6x + 9 = 0$$

Roots: 3, 3

$$a_n = 3^n \cdot 2 + n \cdot 3^n \left(-\frac{2}{3}\right)$$

$$a_0 = 2 \quad a_1 = 4 = 3^n \left[2 + -\frac{2n}{3} \right]$$

$$CE: a_n = 3^n c_1 + n \cdot 3^n c_2$$

$$n=0$$

$$n=1$$

$$a_0 = 3^0 c_1 + 0 \cdot 3^0 c_2 \quad a_1 = 3^1 c_1 + 1 \cdot 3^1 c_2$$

$$2 = c_1$$

$$a_1 = 3c_1 + 3c_2$$

$$a_1 = 6 + 3c_2$$

$$4 = 6 + 3c_2$$

$$3c_2 = 4 - 6$$

$$c_2 = -\frac{2}{3}$$

COMBINATORICS



$$a_n = 8a_{n-1} - 16a_{n-2}, \quad a_0 = 1, \quad a_1 = 3.$$

$$a_n = 4^n - \frac{n}{4} \cdot 4^n.$$

$$= 4^n - n \cdot 4^{n-1}.$$

$$a_n = 8a_{n-1} - 16a_{n-2}$$

$$a_n - 8a_{n-1} + 16a_{n-2} = \textcircled{0} \downarrow$$

COMBINATORICS



Type-4: $a_n = X a_{n-1} + Y a_{n-2} + Z a_{n-3}$.

Roots: r_1, r_2, r_3 .

CE: $a_n = r_1^n c_1 + r_2^n c_2 + r_3^n c_3$.

Type-5:

Roots: r, r, r_1 .

CE: $a_n = r^n c_1 + n r^n c_2 + r_1^n c_3$.

Type-6:

r, r, r .

CE: $a_n = r^n c_1 + n r^n c_2 + n^2 r^n c_3$.

COMBINATORICS



Recurrence Relation

homogenous R.R. :

$$a_n + Xa_{n-1} + Ya_{n-2} = f(n)$$

$$f(n) = 0$$

$$a_n = 8a_{n-1} - 15a_{n-2}$$

$$a_n - 8a_{n-1} + 15a_{n-2} = 0$$

non homogenous R.R.

$$a_n + Xa_{n-1} + Ya_{n-2} = f(n)$$

$$\downarrow$$
$$n$$
$$n^2 + 2$$
$$7^n$$

COMBINATORICS



$$a_n - 3a_{n-1} = 5(7^n)$$

$$a_0 = 2$$

homogeneous

$$a_n - 3a_{n-1} = 0$$

$$x - 3 = 0$$

$$x = 3$$

$$a_n^h = 3^n C$$

$$f(n) = 5(7^n)$$

$$a_n^p = A(7^n)$$

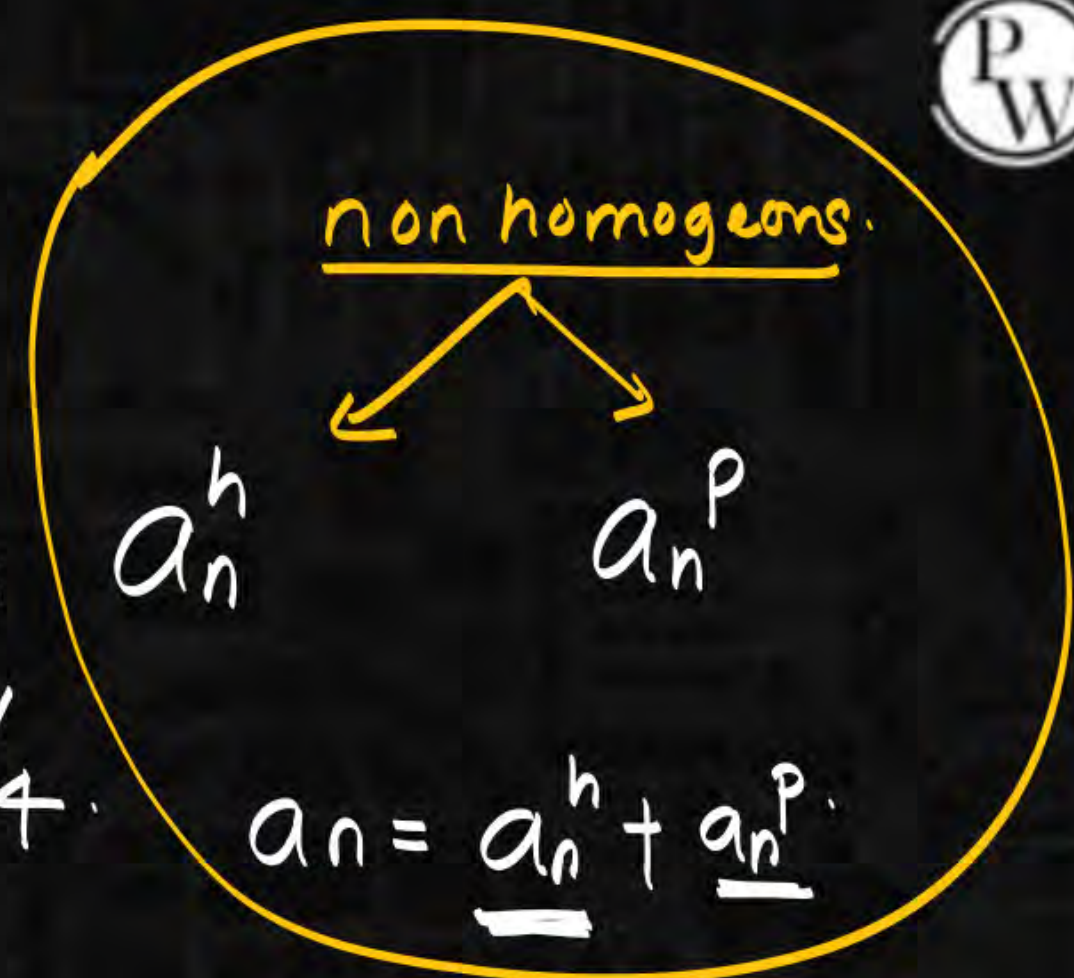
find value of A.

$$\begin{array}{r} A(7^n) - 3A(7^{n-1}) = 5(7^n) \\ \hline \begin{array}{l} \div \text{ by } 7^{n-1} \\ 7^{n-1} \end{array} \end{array}$$

$$7A - 3A = 5 \cdot 7$$

$$4A = 35$$

$$A = 35/4$$



$f(n)$	a_n^p
7^n	$A(7^n)$
3^n	$A(3^n)$
5^n	$A(5^n)$

COMBINATORICS



$$a_0 = 2.$$

$$a_n = a_n^h + a_n^p.$$

$$a_n = 3^n c + A(7^n)$$

$$A = 35/4.$$

$$a_0 = c + \frac{35}{4}.$$

$$2 = c + \frac{35}{4}.$$

$c?$

$$a_n = 3^n c + \frac{35}{4}(7^n)$$

$$c = 2 - \frac{35}{4} = -27/4.$$

$$n=0$$

$$a_0 = 3^0 \cdot c + \frac{35}{4} \cdot 7^0.$$

$$a_n = 3^n \left(-27/4 \right) + \frac{35}{4} (7^n)$$

COMBINATORICS



$$a_n - 3a_{n-1} = 5(3^n) \quad a_0 = 2$$

$$A_n - A_{n+1} + A = 5$$

$$a_n - 3a_{n-1} = 0$$

$$a_n^P = A_n(3^n)$$

$$A = 5$$

$$x - 3 = 0$$

$$x = 3$$

$$a_n^h = 3^n c$$

$$A_n(3^n) - 3(n-1)A3^{n-1} = 5 \cdot 3^n$$

$$\cancel{A_n 3^n} - (n-1)A \cdot \cancel{3^n} = 5 \cancel{3^n}$$

$$a_n = 3^n(2 + 5n)$$

$$\begin{aligned} a_n &= a_n^h + a_n^P \\ &= 3^n c + 5n 3^n \end{aligned}$$

$$a_n = 3^n(c + 5n)$$

$$\begin{aligned} n=0 \\ 2 &= 3^0(c + 5 \cdot 0) \\ c &= 2 \end{aligned}$$

COMBINATORICS



$$a_n - 3a_{n-1} = f(n)$$

Root : 3

$f(n)$

a_n^p

$5(7^n)$

$A(7^n)$

$5(3^n)$

$A_n(3^n)$

$$a_n = 10a_{n-1} - 21a_{n-2} + f(n)$$

$$a_n - 10a_{n-1} + 21a_{n-2} = f(n)$$

$$a_n^n = 7^n c_1 + 3^n c_2$$

Roots : 7, 3

$f(n)$

a_n^p

6^n

$A(6^n)$

3^n

$A_n(3^n)$

COMBINATORICS

$$a_n + \lambda a_{n-1} + \gamma a_{n-2} = f(n)$$

$f(n)$

11.

7^n

n

n^2

a_n^p

A

$A(7^n)$

$An + b$

$An^2 + bn + c$

COMBINATORICS



$$\begin{cases} f_n = f_{n-1} + f_{n-2} \\ f_0 = 0 \quad f_1 = 1 \end{cases}$$

$$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$$

$$\begin{aligned} a_0 &= 2 \\ a_1 &= 5 \\ a_2 &= 15 \end{aligned}$$

$$a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$$

$$a_0 = 1 \quad a_1 = -2 \quad a_2 = -1$$

$$\underline{\text{Sol:}} \quad a_n = (1 + 3n - 2n^2)(-1)^n$$

$$\underline{\text{Ans:}} \quad a_n = 1 - 2^n + 2 \cdot 3^n$$

COMBINATORICS



$$a_{n+1} - a_n = 2n + 3 \quad a_0 = 1.$$

$$a_{n+1} - a_n = 5 \quad a_0 = 3$$

$$a_{n+1} - 2a_n = 2^n \quad a_0 = 1.$$

$$a_{n+2} - 6a_{n+1} + 9a_n = 3(2^n) + 7(3^n)$$

